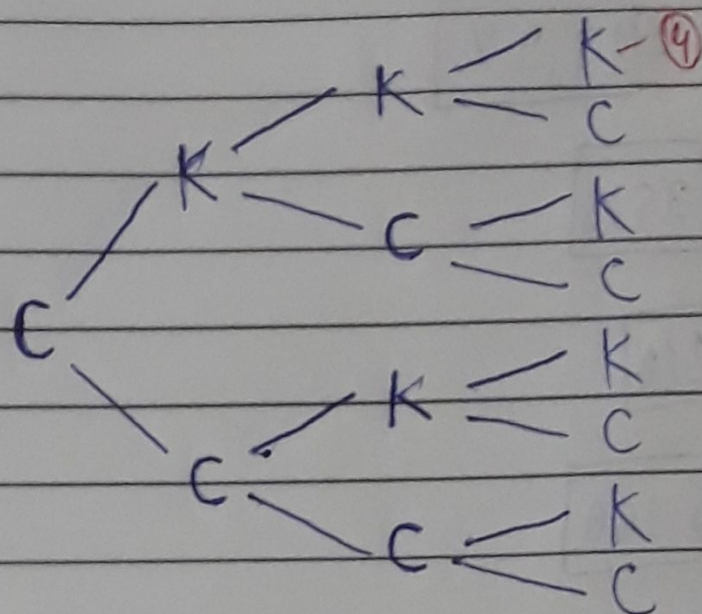


Matricula: 16-95704

Diagram illustrating the construction of a Huffman tree from the frequency distribution:

```
graph LR; K1[K] --- C1[C]; K1 --- C2[C]; C1 --- K2[K]; C1 --- C3[C]; C2 --- K3[K]; C2 --- C4[C]; C3 --- K4[K]; C3 --- C5[C]; C4 --- K5[K]; C4 --- C6[C]; C5 --- K6[K]; C5 --- C7[C]; C6 --- K7[K]; C6 --- C8[C]; C7 --- K8[K]; C7 --- C9[C]; C8 --- K9[K]; C8 --- C10[C]; C9 --- K10[K]; C9 --- C11[C]; C10 --- K11[K]; C10 --- C12[C]; C11 --- K12[K]; C11 --- C13[C]; C12 --- K13[K]; C12 --- C14[C]; C13 --- K14[K]; C13 --- C15[C]; C14 --- K15[K]; C14 --- C16[C]; C15 --- K16[K]; C15 --- C17[C]; C16 --- K17[K]; C16 --- C18[C]; C17 --- K18[K]; C17 --- C19[C]; C18 --- K19[K]; C18 --- C20[C]; C19 --- K20[K]; C19 --- C21[C]; C20 --- K21[K]; C20 --- C22[C]; C21 --- K22[K]; C21 --- C23[C]; C22 --- K23[K]; C22 --- C24[C]; C23 --- K24[K]; C23 --- C25[C]; C24 --- K25[K]; C24 --- C26[C]; C25 --- K26[K]; C25 --- C27[C]; C26 --- K27[K]; C26 --- C28[C]; C27 --- K28[K]; C27 --- C29[C]; C28 --- K29[K]; C28 --- C30[C]; C29 --- K30[K]; C29 --- C31[C]; C30 --- K31[K]; C30 --- C32[C]; C31 --- K32[K]; C31 --- C33[C]; C32 --- K33[K]; C32 --- C34[C]; C33 --- K34[K]; C33 --- C35[C]; C34 --- K35[K]; C34 --- C36[C]; C35 --- K36[K]; C35 --- C37[C]; C36 --- K37[K]; C36 --- C38[C]; C37 --- K38[K]; C37 --- C39[C]; C38 --- K39[K]; C38 --- C40[C]; C39 --- K40[K]; C39 --- C41[C]; C40 --- K41[K]; C40 --- C42[C]; C41 --- K42[K]; C41 --- C43[C]; C42 --- K43[K]; C42 --- C44[C]; C43 --- K44[K]; C43 --- C45[C]; C44 --- K45[K]; C44 --- C46[C]; C45 --- K46[K]; C45 --- C47[C]; C46 --- K47[K]; C46 --- C48[C]; C47 --- K48[K]; C47 --- C49[C]; C48 --- K49[K]; C48 --- C50[C]; C49 --- K50[K]; C49 --- C51[C]; C50 --- K51[K]; C50 --- C52[C]; C51 --- K52[K]; C51 --- C53[C]; C52 --- K53[K]; C52 --- C54[C]; C53 --- K54[K]; C53 --- C55[C]; C54 --- K55[K]; C54 --- C56[C]; C55 --- K56[K]; C55 --- C57[C]; C56 --- K57[K]; C56 --- C58[C]; C57 --- K58[K]; C57 --- C59[C]; C58 --- K59[K]; C58 --- C60[C]; C59 --- K60[K]; C59 --- C61[C]; C60 --- K61[K]; C60 --- C62[C]; C61 --- K62[K]; C61 --- C63[C]; C62 --- K63[K]; C62 --- C64[C]; C63 --- K64[K]; C63 --- C65[C]; C64 --- K65[K]; C64 --- C66[C]; C65 --- K66[K]; C65 --- C67[C]; C66 --- K67[K]; C66 --- C68[C]; C67 --- K68[K]; C67 --- C69[C]; C68 --- K69[K]; C68 --- C70[C]; C69 --- K70[K]; C69 --- C71[C]; C70 --- K71[K]; C70 --- C72[C]; C71 --- K72[K]; C71 --- C73[C]; C72 --- K73[K]; C72 --- C74[C]; C73 --- K74[K]; C73 --- C75[C]; C74 --- K75[K]; C74 --- C76[C]; C75 --- K76[K]; C75 --- C77[C]; C76 --- K77[K]; C76 --- C78[C]; C77 --- K78[K]; C77 --- C79[C]; C78 --- K79[K]; C78 --- C80[C]; C79 --- K80[K]; C79 --- C81[C]; C80 --- K81[K]; C80 --- C82[C]; C81 --- K82[K]; C81 --- C83[C]; C82 --- K83[K]; C82 --- C84[C]; C83 --- K84[K]; C83 --- C85[C]; C84 --- K85[K]; C84 --- C86[C]; C85 --- K86[K]; C85 --- C87[C]; C86 --- K87[K]; C86 --- C88[C]; C87 --- K88[K]; C87 --- C89[C]; C88 --- K89[K]; C88 --- C90[C]; C89 --- K90[K]; C89 --- C91[C]; C90 --- K91[K]; C90 --- C92[C]; C91 --- K92[K]; C91 --- C93[C]; C92 --- K93[K]; C92 --- C94[C]; C93 --- K94[K]; C93 --- C95[C]; C94 --- K95[K]; C94 --- C96[C]; C95 --- K96[K]; C95 --- C97[C]; C96 --- K97[K]; C96 --- C98[C]; C97 --- K98[K]; C97 --- C99[C]; C98 --- K99[K]; C98 --- C100[C]; C99 --- K100[K]; C99 --- C101[C]; C100 --- K101[K]; C100 --- C102[C]; C101 --- K102[K]; C101 --- C103[C]; C102 --- K103[K]; C102 --- C104[C]; C103 --- K104[K]; C103 --- C105[C]; C104 --- K105[K]; C104 --- C106[C]; C105 --- K106[K]; C105 --- C107[C]; C106 --- K107[K]; C106 --- C108[C]; C107 --- K108[K]; C107 --- C109[C]; C108 --- K109[K]; C108 --- C110[C]; C109 --- K110[K]; C109 --- C111[C]; C110 --- K111[K]; C110 --- C112[C]; C111 --- K112[K]; C111 --- C113[C]; C112 --- K113[K]; C112 --- C114[C]; C113 --- K114[K]; C113 --- C115[C]; C114 --- K115[K]; C114 --- C116[C]; C115 --- K116[K]; C115 --- C117[C]; C116 --- K117[K]; C116 --- C118[C]; C117 --- K118[K]; C117 --- C119[C]; C118 --- K119[K]; C118 --- C120[C]; C119 --- K120[K]; C119 --- C121[C]; C120 --- K121[K]; C120 --- C122[C]; C121 --- K122[K]; C121 --- C123[C]; C122 --- K123[K]; C122 --- C124[C]; C123 --- K124[K]; C123 --- C125[C]; C124 --- K125[K]; C124 --- C126[C]; C125 --- K126[K]; C125 --- C127[C]; C126 --- K127[K]; C126 --- C128[C]; C127 --- K128[K]; C127 --- C129[C]; C128 --- K129[K]; C128 --- C130[C]; C129 --- K130[K]; C129 --- C131[C]; C130 --- K131[K]; C130 --- C132[C]; C131 --- K132[K]; C131 --- C133[C]; C132 --- K133[K]; C132 --- C134[C]; C133 --- K134[K]; C133 --- C135[C]; C134 --- K135[K]; C134 --- C136[C]; C135 --- K136[K]; C135 --- C137[C]; C136 --- K137[K]; C136 --- C138[C]; C137 --- K138[K]; C137 --- C139[C]; C138 --- K139[K]; C138 --- C140[C]; C139 --- K140[K]; C139 --- C141[C]; C140 --- K141[K]; C140 --- C142[C]; C141 --- K142[K]; C141 --- C143[C]; C142 --- K143[K]; C142 --- C144[C]; C143 --- K144[K]; C143 --- C145[C]; C144 --- K145[K]; C144 --- C146[C]; C145 --- K146[K]; C145 --- C147[C]; C146 --- K147[K]; C146 --- C148[C]; C147 --- K148[K]; C147 --- C149[C]; C148 --- K149[K]; C148 --- C150[C]; C149 --- K150[K]; C149 --- C151[C]; C150 --- K151[K]; C150 --- C152[C]; C151 --- K152[K]; 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```



2 → $K_{\text{space control}} = 6 \cdot 6 = 36$

$4 = 1+3$ or $2+2$ or $3+1$

$$5 = 1+4 \text{ or } 2+3 \text{ or } 3+2 \text{ or } 4+1$$

Evento = 7

3-7) Esporo anastrol = 12
Evento = 3

4 → Espaco amostral = $2^3 = 8$
Eventos = 3

$$17 \rightarrow P = 2058 \div 6485 \approx 0,32$$

$$18 \rightarrow P = \frac{407}{4834} \approx 0,08$$



Nome: Marcos Vinícius da Silva

Matrícula: 26-95904

$$35 \rightarrow 24 + 3,9 = 27,9\%$$

$$36 \rightarrow 21 + 14 = 35\%$$

$$41 \rightarrow A_{11,2} = 210$$

$$42 \rightarrow A_{8,6} = 20.260$$

$$43 \rightarrow C_{9,4} = 35$$

$$44 \rightarrow \frac{C_{5,3}}{C_{10,3}} \approx 0,08\bar{3}$$

$$50 \rightarrow 23 \cdot 26 \cdot 26 \cdot 20 = 255.480$$

$$P = \frac{1}{255480} \approx 6,43 \cdot 10^{-6}$$

$$51 \rightarrow P = \frac{199}{200} \cdot \frac{196}{199} \cdot \frac{195}{198} \approx 0,95$$

$$b) P = \frac{3}{200} \cdot \frac{2}{199} \cdot \frac{1}{198} \approx 7,62 \cdot 10^{-9}$$

$$c) P \approx 1 - 0,95 \approx 0,049$$

$$d) P \approx 1 - 7,62 \cdot 10^{-9} \approx 0,999$$

$$52 \rightarrow \infty \frac{346}{350} \cdot \frac{345}{349} \cdot \frac{344}{348} \cdot \frac{343}{347} \approx 0,95$$

Nome: Marcos Vinícius da Silva

Matrícula: 26-95704

$$b) \frac{4}{350} \cdot \frac{3}{349} \cdot \frac{2}{348} \cdot \frac{1}{347} \approx \boxed{1,63 \cdot 10^{-9}}$$

$$c) \frac{4}{350} \cdot \frac{3}{349} \approx \boxed{9,82 \cdot 10^{-5}}$$

$$d) 1 - 0,95 \approx \boxed{0,042}$$

$$19 \rightarrow P(0) = \frac{26}{2600} = \boxed{0,01}$$

$$P(2) = \frac{442}{2600} = \boxed{0,17}$$

$$P(2) = \frac{928}{2600} = \boxed{0,28}$$

$$P(3) = \frac{2404}{2600} = \boxed{0,54}$$

$$20 \rightarrow P(0) = \frac{6}{192} \approx \boxed{0,03}$$

$$P(2) = \frac{12}{192} \approx \boxed{0,06}$$

$$P(2) = \frac{29}{192} \approx \boxed{0,15}$$

$$P(3) = \frac{59}{192} \approx \boxed{0,3}$$

$$P(4) = \frac{42}{192} \approx 0,22$$

$$P(5) = \frac{30}{192} \approx 0,16$$

$$P(6) = \frac{26}{192} \approx \boxed{0,083}$$